

Problem 1: Your friend shows you an algorithm INV that can invert a given function f in PPT, but only for those $f(x)$ where the first 20 high-order bits of x are all 1's. However, if the first 20 bits are not all 1's, you assume that for any PPT adversary A , the probability that A inverts F becomes negligible as the length of the input increases. Could f be a one-way function? (Prove your answer). If not, is there a way to use f as a building block to build a one-way function? If so, show how to do it and prove your answer; if not, show a counterexample.

Ans: The function is not one way. This is because, for a constant fraction ($1/2^{20}$), we can invert f in PPT. We can use f as a building block for a one way function in the following manner by constructing function f' . On input x , we can return $f(x) \oplus f(\hat{x})$. Only one of the two outputs can have the first 20 bits as 1, so

Problem 6. Consider a secure pseudo-random generator (PRG) G that maps n -bit seed s to $2n$ -bit output $G(s)$. Define $LH(*)$ to be a function that takes as an input $2n$ bits, and outputs the first half of its input bits. Define $RH(*)$ similarly (for the right half). Define:

$$F(s) = G(LH(G(s)))G(RH(G(s)))$$

1. The output length is $4n$.
2. We demonstrate the security of G by reduction, showing that an adversary that can distinguish a random number from $F(s)$ can differentiate a random number from $G(s)$ with non-negligible probability. This is done using the hybrid argument. We demonstrate that if we have an output that can differentiate between an output of 4 random n -bit long numbers and $G(s)$, G is not a secure PRG. Firstly, this implies the existence of an adversary A that outputs 1 if the input is $G(s)$ with non-negligible probability. The idea is the following. First, we replace the left n bits of $G(s)$ with a truly random number. Therefore, the final output of this hybrid machine will be:

$$F_1(S) = G(LH(r_l))G(RH(G(s)))$$

(a) First